

DealFlow Ai

PART ONE: INTRO TO DEALFLOW AI

Q: WHAT IS IT?

Dealflow Ai is going to be both a web and mobile based application - after we nail the web application we will start working on the mobile version. There are several moving parts that make up this “**Off Market Deal Finder**”, but essentially works on behalf of business brokers and other individuals that occupy an industry along that same vertical. We will be employing an ensemble machine learning module set with a very specific 55+ weighted scoring metric that it will apply against data points ingested from around a dozen different sources via API. All data ingested will form entries in our data base where a 3 stage feedback loop will feed the action/outcome of the scored leads back into the ensemble to keep the scoring metric sharp and continue to pump out accurate off market leads. This is a subscription based service with 2 Subscription Tiers to choose from. Each Tier pays a set up fee and a monthly retainer. When we receive organic leads from either visits to our front end website or via our social media platform channels, the user signs up, pays and goes through onboarding to obtain access to their custom dashboard and they will then have certain capabilities dependent upon the Tier Subscription they are signed up for. The goal of the system is to provide a better and more accurate way to source off market leads for businesses showing signs or markers of an entity that might end up for sale by owner. The business brokers today are finding they are struggling with having an overage of buyers and not enough sellers- unfortunately owners make the mistake of letting their business go when falling on hard times rather than selling it into the marketplace. The same goes for when a business owner passes away and leaves their entity to a beneficiary who is not willing to take it on and lets it go rather than attempting to run it through the marketplace. So we are attempting to catch these businesses before they fall at the first sign of failure and match these brokers with owners and get a contract made.

Q: WHO IS IT FOR:

As I have stated, the platform is geared towards Business Brokers looking for those off market deals by using weighted metrics to score leads and matching them with potential sellers before the they hit the market or the owner even knows they are ready to sell. More often than not the owner starts searching for solutions casually when they are wanting to exit or are planning for an exit in the near future. If we can lock down an

accuracy rate of 85% with our Business Brokers we can then expand our reach into the adjacent industries such as commercial and residential real estate, insurance professionals, estate and trust firms including probate attorney's, accountants and private family offices. We may open up to investors later on down the road but for the foreseeable future we are just hoping to make capturing potential leads for our business broker clients immeasurably easier and at the same time mitigating the tremendous loss of revenue producing small and medium sized entities within our communities and even spanning outward to reach the multitude of enterprise level afflicted organizations here in the United States.

Multi-Touch Outreach Campaign

- Cold call script focusing on "confidential valuation" not "selling"
- LinkedIn InMail to business owners (even if they barely use it, messages get email notifications)
- Physical mail that doesn't look like marketing - handwritten envelopes, insider reports
- Text message sequences to cell phones (compliance-checked)
- All happening simultaneously, not some weak drip campaign

Infiltrate Where Owners Actually Are

- Country clubs and golf courses (sponsor a hole, get the member list)
- Industry trade associations (plumbing, HVAC, manufacturing groups)
- Rotary, Lions, Kiwanis - where old-school owners network
- Estate planning attorney referral partnerships

Competitive Intelligence

- Scrape every competitor's listings
- Contact their sellers who've been listed 6+ months
- "Your current broker has had 6 months. Let's discuss alternative strategies"
- Clay.com or Apollo.io for data enrichment
- Twilio/TextMagic for compliant SMS outreach
- Instantly or Smartlead for email sequences that don't get spam-filtered
- PhantomBuster for LinkedIn automation
- Aircall or JustCall for power dialing with local presence

The Referral Network That Actually Converts

Not "tell your friends" but structured partnerships:

- CPAs get \$5K for successful deal referrals
- Estate attorneys get reciprocal referrals

- Commercial real estate brokers get bird dog fees
- Wealth managers get introduced to sellers post-close

Voice AI Qualifier

- Outbound calls presenting as "business valuation research firm"
- Natural conversation flow that extracts timeline, motivation, financials
- Books qualified callbacks directly to calendar
- Records calls for sentiment analysis and coaching

Intelligence Aggregator

- Your data feeds into enrichment pipeline
- Also Pulls social media, news mentions, litigation records
- Builds psychological profile and approach strategy per lead
- Predicts likelihood to sell score

Multichannel Orchestration

- SMS with local presence number
- LinkedIn via phantom profiles
- Personalized video messages via Sendspark
- Physical mail timed with digital touches
- All tracked in unified campaign dashboard

The Trojan Horse Approach

Position this as "Phoenix Business Intelligence Platform" not "lead generation." Steve gets:

- Weekly intelligence reports on businesses likely to sell
- Pre-call briefs with owner psychology profiles
- Conversation scripts based on trigger events
- Automated follow-up sequences

ABOVE: 🙌 This what sparked the original idea, but some aspects kind of got lost in the proverbial sauce and I want to make sure I'm still integrating all the magic and great ideas here with what I have laid/written out **BELOW:** 🙌

ALSO: I'm not 100% in love with the metrics and the weight system for all the data points that are laid out below. I would love it if you could do it overhaul in this entire system to make it incredibly fucking accurate on point and up-to-date.

THANK YOU!!

SYSTEM SO FAR

LEAD DISTRIBUTION SYSTEM:

New Lead Distribution Model (No Regional Restrictions):

- Open marketplace - all brokers can access the platform with paid subscription
- Leads are randomly distributed across all active brokers according to their predefined data points given at sign up/onboarding (i.e. broker says- I only want businesses in the greater phoenix area, ARR of \$2 million+, at least 3-5 employees and I DO NOT WANT anything in the restaurant/food service industries)
- Black tier leads are not assigned. They are monitored closely for any changes after rescoring as are the rest of the businesses within our date in Vegas however, these are a little bit higher priorities since they have already received a score on our index.
- No territorial exclusivity (can be assigned leads for any geographical location they define, however, once a lead is assigned to a broker it stays with them for the duration of active status within our system- leads can be clawed back if mis-assigned by admin on admin dashboard OR a lead can be re-assigned to another user after original disbursement if the original user leaves our platform and no contact had been made- at that point the lead gets put back in the color pool and distributed to the next user in line the next disbursement day)
- Administrators are the only one who can manually revoke/override lead assignments and reassign them when necessary.
- No broker caps per region
- Mix of Platinum/Gold/Silver distributed to keep pipeline balanced (after the lead is scored and given a natural language explanation for the scoring it is placed into its designated color bucket for outreach and dashboard disbursement- color bucket label is chosen based on the cumulative score given by the ML module assessing the data points based on the outlined weights in the ML document).

ROUND ROBIN LEAD DISTRIBUTION

SUBSCRIPTION TIER ONE:

- Each broker gets their 60 leads/month (20 of each color)

- Distribution happens weekly at 6 AM every Tuesday morning (15/Week, 5/Each Color)
- Algorithm ensures no broker gets the same lead
- Distribution happens via Chronjob
- As soon as a lead hits the user (broker) dashboard, the lead receives an **INITIAL** complimentary “warm email” with value adding attachments from the brokers document vault, personalized video message and a calendar scheduling link for said broker. After that its up to the broker to follow up, make contact and provide an outcome action that gets fed into the feedback loop that retrains the ML Module so that it keeps improving over time. The lead is also sent a USPS postcard to both of the business address and the home address of the owner(s).
- This **INITIAL** outreach is tracked for engagement and our AI automation system will provide outcome actions for these automatic touches.
- If at any point a lead schedules using the magic link in the outreach email our system flags at with an outcome as the system itself will send that automatic confirmation email to both parties scheduled and the lead be placed on the brokers calendar per the norm.
- The **FULL** cycle Ai Outreach Automation is something that is only included in Tier 2 of the subscription offerings (costs more monthly and can be up/downgraded at any time from the users account settings on the users dashboard)

SUBSCRIPTION TIER TWO:

- Each broker gets their 60 leads/month (20 of each color)
- Distribution happens weekly (15/Week, 5/Each Color)
- Algorithm ensures no broker gets the same lead
- Distribution happens every Tuesday at 6am via Chronjob
- As soon as a lead hits the user (broker) dashboard, the lead receives an **INITIAL** complimentary “warm email” with value adding attachments from the brokers document vault, personalized video message and a calendar scheduling link for said broker. After that its up to the broker to follow up, make contact and provide an outcome action that gets fed into the feedback loop that retrains the ML Module so that it keeps improving over time. The lead is also sent a USPS postcard to both of the business address and the home address of the owner(s).

- At this point, the lead qualification and outreach system is deployed (SEE BELOW SECTION ON AI: CONCIERGE, MODULE AND LEAD QUALIFICATION SYSTEM).

AI CALLING CONCIERGE (LEAD QUALIFICATION/OUTREACH SYSTEM)

OUTREACH METHODS & SCHEDULE:

1. **As discussed above, the initial outreach is mandatory for both tiers however any subsequent outreach and follow up it's only available on tier 2 subscriptions.**
2. **Initial 1st Touch Warm Outreach Email:** Sent on **Day 1 @ 6am** after disbursement:
 - Using ChronJobs and Inngest the 1st touch email is sent immediately upon disbursement to the Brokers dashboard. It includes value add attachments from the Brokers document vault (dashboard tab), a personalized Ai generated introduction video and a calendar booking link tied to said Brokers schedule & adding the appointment/booking details to Broker calendar
 - Any bookings made whether it's via the link in the email or via telephone by any user or Broker themselves, a confirmation email and/or SMS Message is generated and sent out to all parties on the booking.
 - Engagement Tracker is deployed and monitors activity then if lead books via link from Warm Intro Email auto-confirm for booking occurs and there is also a "Reject" or "DNC" button on the email itself in case the lead wishes to not be contacted again. If DNC is used the lead is actioned and deactivated in our system then removed from the Brokers dashboard.
3. **Direct Mail Postcard Sent To Both The Business Address & Owner Home Address Via Usps Using The Pre-Designed Template Created In Canva And The Existing Indica For Payment**
 - This occurs on Day 1 @ 6am, runs on autopilot using the same mechanism as above and initiates on lead disbursement to dashboard

4. 1st Follow Up Phone Call - 72 Hours after Lead Disbursement (Day 3)

- Ai Voice Agent Autodials each lead outstanding for every Tier 2 Subscription Broker.
- Agent attempts phone call a minimum of 3 and maximum of 5 times in the 8 hours allotted (between 10am and 6pm)
- After each attempt Ai Agent is required to “Action” the outcome for each attempt for the feedback loop.
- If agent makes contact and is able to schedule the lead on the Brokers calendar (Booking is Main Goal of Outreach Attempts), Agent actions the lead for feedback loop and the usual confirmation email/sms mechanism is initiated from the booking.
- Lead “Action” follows feedback loop funnel (see 3-Stage Feedback Loop)

5. 2nd Follow Up Phone Call - 48 Hours after 1st Follow Up Phone Call (Day 5)

- Follows same rules outlined in bullet point #3 above
- if no contact is made via telephone proceed to bullet point #6

6. 3rd Follow Up Phone Call - 48 Hours after 2nd Follow Up Phone Call (Day 7)

- Follows same rules outlined in bullet point #3 above
- If no contact is made via telephony the agent is to immediately proceed to bullet point #7 and send out that Second follow up email. This must be done same day.

7. 2nd Follow Up Email - Immediately After 3rd Follow Up Phone Call (bullet point#6)

- Uses template in email series
- Follows same rules outlined in bullet point #3 above
- If no contact is made Follow Up Phone Calls continue every 48 hours for the next 21 days - simultaneously and running concurrent to the phone calls, the Follow Up Emails are initiated once a week for the next 3 weeks.
- At day 30 after the lead has been disbursed to the Brokers dashboard, if no contact has been made during the cycle made we begin a “14 Day Pause” and then repeat the full Follow Up Cycle again until we either

Speak to the lead and get the booking on the calendar or they tell us to not contact them again and we de-activate the lead in our database.

PART TWO: DEALFLOW AI- COMPLETE ML ARCHITECTURE FROM SCRATCH

WHAT THIS SECTION CONTAINS

1. **Complete Data Source Mapping:** Every Dewey dataset + your owned APIs → exactly what signals each provides
 2. **Full Database Schema:** 20+ tables capturing businesses, owners, all distress signals, trends, and outcomes
 3. **55+ Weighted Features:** Organized into 7 categories with specific point values and threshold logic
 4. **Ensemble Model Design:** XGBoost (42%) + Random Forest (33%) + Neural Net (25%) with configs
 5. **Tier System:** Platinum (75-100, 0-3mo) → Gold (55-74, 3-6mo) → Silver (35-54, 6-12mo) → Unqualified (<35)
 6. **LLM Explanation Layer:** Natural language reasoning for each scored lead
 7. **Three-Stage Feedback Loop:** Outreach → Meeting → Deal Closure, all feeding back into training
 8. **Event-Driven Rescoring:** Automatic re-scoring when critical signals fire
-

PART 1: COMPLETE DATA SOURCE INVENTORY

Every single source I have access to, what it provides, and how it maps to signals (Need to also map database columns and rows to each data source and obtain API endpoints for wiring up ingestion layers)

DEWEY DATA PLATFORM (Your Primary)

Dataset	Records	What It Gives You
People Data Labs - Company Insights	72M companies	Business name, address, NAICS/SIC, employee count, revenue estimates, founded date, website, phone
People Data Labs - Employee Count by Month	72M	Historical headcount trends (12-24 months)
People Data Labs - Recent Executive Departures	—	C-suite/VP departures with dates
People Data Labs - Person Demographics	2.4B	Owner age, education, income estimates, interests, location
People Data Labs - Skills	2.4B	Self-reported skills
People Data Labs - Interest	308M	Self-reported interests
People Data Labs - Job Title Level	132M	Seniority level of individuals
People Data Labs - Experience Title Level	420M	Experience/job levels
People Data Labs - Location Name	748M	Current address of individuals
ATTOM Pre-Foreclosure History	27M	Pre-foreclosure notices, notice of default dates, foreclosure sale dates
ATTOM Tax Assessor	158M properties	Assessed values, tax delinquency flags, property characteristics
ATTOM Assessor History	2.7B	YoY valuation changes, tax payment history
ATTOM Recorder	578M	Deed transfers, mortgage recordings, lien filings, UCC statements
ATTOM AVM	98M	Property valuations, confidence scores, value trends
Built Building Permits	178M	Permit issue dates, types, costs — identifies deferred maintenance

SafeGraph Spend Patterns	82M	Credit/debit transaction data by POI
SafeGraph Global Places (POI) & Geometry	60M	Points of interest, building footprints
Advan Foot Traffic / Monthly Patterns	1B	Visitor counts, demographics, dwell time by POI
Verisk Property Data	84M	Property location, ownership, year built, building size, condition
BrightQuery Sector	300	Financials & employment in time series, SBA loans
WageScape Job Postings	423M	Job postings with salary data — declining postings = contraction
RentHub Rental Data	408M	Rental listings, pricing, property info
ConsumerEdge Daily Spend	400M	Aggregated consumer transaction data by brand/state
REsimplifi Commercial Listings	—	Broker data, suite listings
ClimateCheck U.S. Climate Risk	33K	Hazard ratings (relevant for property risk)
Context Analytics - Twitter Sentiment	9.8M	Social sentiment for equities

MY ADDITIONAL SUBSCRIPTIONS / APIs TO USE OR BUILD FOR DATA INGESTION

Source	What It Gives You
Apollo.io (18.2M records) Dewey*	Decision-maker names, emails, LinkedIn profiles, direct dials & Owner age/DOB, Current/Historical Address/Phone/Email, Relatives and Associates
UniCourt API (Free)	Bankruptcy filings (Ch. 7, 11, 13), civil judgments, divorce records, liens, business litigation, partnership dissolutions
Dun & Bradstreet Scraper (Free)	PAYDEX score (payment performance), Failure Score, Days Beyond Terms, credit utilization, payment trends

SEC Edgar (Free)	10-K, 10-Q, 8-K filings, Form 4 insider trading, going concern warnings, bankruptcy filings
Google APIs (Free)	Google Places (ratings, reviews, photos), Google My Business data
NASDAQ API (Free)	Industry benchmarks, revenue multiples, margins by sector
Yelp API (Free tier)	Business ratings, review counts, recent reviews for trend analysis

ADDITIONAL SIGNALS TO BUILD/SCRAPE

Source	What It Gives You
Custom Facebook Scraper	Business page activity, post frequency, engagement decay
Custom Instagram Scraper	Post frequency, follower trends, engagement rates
Custom LinkedIn Scraper	Company page activity, job postings (hiring vs. stagnation)
State SOS Databases	Entity status (active/dissolved/suspended), formation date, registered agent changes
County Recorder Offices	UCC filings, mechanic liens, judgment liens
Obituaries/Death Notices	Online filing of Obituaries from funeral homes or death notices In local newspapers (online version)

PART 2: DATABASE SCHEMA FOR RAW DATA INGESTION

This is the foundation. Every business gets a profile. Every signal gets tracked with timestamps.

```
-- =====
-- CORE BUSINESS ENTITY (The Master Record)
-- =====
CREATE TABLE businesses (
  id UUID PRIMARY KEY DEFAULT gen_random_uuid(),
```

```

-- Identity
pdl_company_id VARCHAR(100) UNIQUE,      -- People Data Labs ID (primary key for
matching)
duns_number VARCHAR(9),                  -- D&B identifier
ein VARCHAR(10),                        -- Federal EIN if available

-- Basic Info
legal_name VARCHAR(255) NOT NULL,
dba_name VARCHAR(255),
business_type VARCHAR(50),              -- LLC, Corp, Partnership, Sole Prop
naics_code VARCHAR(6),
sic_code VARCHAR(4),
industry_category VARCHAR(100),

-- Location
street_address VARCHAR(255),
city VARCHAR(100),
state VARCHAR(2),
zip_code VARCHAR(10),
county VARCHAR(100),
latitude DECIMAL(10, 7),
longitude DECIMAL(10, 7),

-- Business Metrics (from PDL + D&B)
founded_date DATE,
years_in_business INTEGER GENERATED ALWAYS AS (
    EXTRACT(YEAR FROM CURRENT_DATE) - EXTRACT(YEAR FROM founded_date)
) STORED,
employee_count INTEGER,
employee_count_range VARCHAR(20),        -- '1-10', '11-50', '51-200', etc.
revenue_estimate DECIMAL(15, 2),
revenue_range VARCHAR(30),              -- '<$1M', '$1-5M', '$5-10M', etc.

-- Contact
main_phone VARCHAR(20),
website VARCHAR(255),

-- Status
entity_status VARCHAR(20) DEFAULT 'active', -- active, dissolved, suspended
data_quality_score DECIMAL(3, 2),        -- 0.00-1.00 confidence in data

-- Timestamps
created_at TIMESTAMPTZ DEFAULT NOW(),

```

```

updated_at TIMESTAMPTZ DEFAULT NOW(),
last_enriched_at TIMESTAMPTZ,

-- Indexes for fast lookups
CONSTRAINT idx_state_industry UNIQUE (state, naics_code, id)
);

CREATE INDEX idx_businesses_location ON businesses(state, city);
CREATE INDEX idx_businesses_industry ON businesses(naics_code);
CREATE INDEX idx_businesses_revenue ON businesses(revenue_estimate);
CREATE INDEX idx_businesses_age ON businesses(years_in_business);

-- =====
-- BUSINESS OWNER (One-to-Many: Business can have multiple owners)
-- =====
CREATE TABLE business_owners (
  id UUID PRIMARY KEY DEFAULT gen_random_uuid(),
  business_id UUID REFERENCES businesses(id) ON DELETE CASCADE,

  -- Identity
  pdl_person_id VARCHAR(100),          -- People Data Labs Person ID

  -- Basic Info
  first_name VARCHAR(100),
  last_name VARCHAR(100),
  full_name VARCHAR(255),

  -- Demographics (Critical for scoring)
  date_of_birth DATE,
  age INTEGER GENERATED ALWAYS AS (
    EXTRACT(YEAR FROM AGE(CURRENT_DATE, date_of_birth))
  ) STORED,
  gender VARCHAR(20),

  -- Contact
  personal_email VARCHAR(255),
  business_email VARCHAR(255),
  mobile_phone VARCHAR(20),
  linkedin_url VARCHAR(500),

  -- Address
  home_street VARCHAR(255),
  home_city VARCHAR(100),
  home_state VARCHAR(2),

```

```

home_zip VARCHAR(10),

-- Ownership
ownership_percentage DECIMAL(5, 2),
role VARCHAR(100),           -- Owner, Partner, CEO, President
is_primary_owner BOOLEAN DEFAULT FALSE,
ownership_start_date DATE,

-- Timestamps
created_at TIMESTAMPTZ DEFAULT NOW(),
updated_at TIMESTAMPTZ DEFAULT NOW(),

UNIQUE(business_id, pdl_person_id)
);

CREATE INDEX idx_owners_age ON business_owners(age);
CREATE INDEX idx_owners_business ON business_owners(business_id);

-- =====
-- OWNER LIFE EVENTS (Divorce, Death, Health - from UniCourt + Public Records)
-- =====
CREATE TABLE owner_life_events (
  id UUID PRIMARY KEY DEFAULT gen_random_uuid(),
  owner_id UUID REFERENCES business_owners(id) ON DELETE CASCADE,

  event_type VARCHAR(50) NOT NULL,           -- 'divorce', 'spouse_death', 'health_issue',
  'partnership_dissolution'
  event_date DATE,
  event_source VARCHAR(100),                 -- 'unicourt', 'obituary_scrape', 'court_record'

-- Event Details
case_number VARCHAR(100),
court_name VARCHAR(255),
jurisdiction VARCHAR(100),

-- Status
event_status VARCHAR(50),                   -- 'filed', 'pending', 'finalized'

-- Raw Data
raw_record JSONB,

created_at TIMESTAMPTZ DEFAULT NOW(),

UNIQUE(owner_id, event_type, event_date)

```

```
);
```

```
CREATE INDEX idx_life_events_type ON owner_life_events(event_type);
CREATE INDEX idx_life_events_date ON owner_life_events(event_date DESC);
```

```
-- =====
```

```
-- OWNER FINANCIAL DISTRESS (Personal Bankruptcies, Liens, Judgments)
```

```
-- =====
```

```
CREATE TABLE owner_financial_distress (
  id UUID PRIMARY KEY DEFAULT gen_random_uuid(),
  owner_id UUID REFERENCES business_owners(id) ON DELETE CASCADE,

  distress_type VARCHAR(50) NOT NULL,          -- 'bankruptcy_ch7', 'bankruptcy_ch13',
  'tax_lien', 'judgment', 'mechanics_lien'

  -- Filing Details
  filing_date DATE,
  case_number VARCHAR(100),
  court_name VARCHAR(255),
  jurisdiction VARCHAR(100),

  -- Amounts
  amount DECIMAL(15, 2),

  -- Status
  status VARCHAR(50),                          -- 'filed', 'active', 'discharged', 'dismissed', 'satisfied'
  discharge_date DATE,

  -- Source
  data_source VARCHAR(100),                     -- 'unicourt', 'county_recorder', 'data_com'
  raw_record JSONB,

  created_at TIMESTAMPTZ DEFAULT NOW(),

  UNIQUE(owner_id, distress_type, case_number)
);
```

```
CREATE INDEX idx_owner_distress_type ON owner_financial_distress(distress_type);
CREATE INDEX idx_owner_distress_date ON owner_financial_distress(filing_date DESC);
```

```
-- =====
```

```
-- BUSINESS FINANCIAL DISTRESS (Pre-foreclosure, Tax Delinquency, Business BK)
```

```
-- =====
```

```
CREATE TABLE business_financial_distress (
```

```

id UUID PRIMARY KEY DEFAULT gen_random_uuid(),
business_id UUID REFERENCES businesses(id) ON DELETE CASCADE,

distress_type VARCHAR(50) NOT NULL,          -- 'pre_foreclosure', 'tax_delinquent',
'bankruptcy_ch11', 'ucc_filing', 'judgment_lien'

-- Property Link (if applicable)
property_apn VARCHAR(50),
property_address VARCHAR(255),

-- Filing Details
filing_date DATE,
notice_date DATE,
auction_date DATE,
case_number VARCHAR(100),

-- Amounts
amount_owed DECIMAL(15, 2),
original_loan_amount DECIMAL(15, 2),

-- Status
status VARCHAR(50),                          -- 'notice_filed', 'auction_scheduled', 'sold',
'redeemed', 'active', 'satisfied'

-- Lender/Creditor
creditor_name VARCHAR(255),

-- Source
data_source VARCHAR(100),                     -- 'attom', 'unicourt', 'county_recorder'
raw_record JSONB,

created_at TIMESTAMPTZ DEFAULT NOW(),

UNIQUE(business_id, distress_type, case_number)
);

CREATE INDEX idx_biz_distress_type ON business_financial_distress(distress_type);
CREATE INDEX idx_biz_distress_date ON business_financial_distress(filing_date DESC);

-- =====
-- BUSINESS CREDIT PROFILE (D&B Data)
-- =====
CREATE TABLE business_credit (
  id UUID PRIMARY KEY DEFAULT gen_random_uuid(),

```



```

business_id UUID REFERENCES businesses(id) ON DELETE CASCADE,

-- D&B Scores
paydex_score INTEGER,           -- 1-100, <50 = distress signal
failure_score INTEGER,         -- Higher = more likely to fail, >1800 = danger
delinquency_score INTEGER,

-- Payment Behavior
days_beyond_terms INTEGER,     -- Average days late on payments
payment_trend VARCHAR(20),     -- 'improving', 'stable', 'declining'

-- Credit Utilization
credit_limit_used_pct DECIMAL(5, 2),

-- Trade References
trade_references_count INTEGER,
satisfactory_trade_pct DECIMAL(5, 2),

-- Snapshot Date
snapshot_date DATE NOT NULL,
data_source VARCHAR(50) DEFAULT 'dnb_scraper',
raw_record JSONB,

created_at TIMESTAMPTZ DEFAULT NOW(),

UNIQUE(business_id, snapshot_date)
);

CREATE INDEX idx_credit_paydex ON business_credit(paydex_score);
CREATE INDEX idx_credit_failure ON business_credit(failure_score);

-- =====
-- EMPLOYEE TRENDS (PDL Employee Count by Month)
-- =====
CREATE TABLE employee_trends (
  id UUID PRIMARY KEY DEFAULT gen_random_uuid(),
  business_id UUID REFERENCES businesses(id) ON DELETE CASCADE,

  snapshot_month DATE NOT NULL,      -- First of month
  employee_count INTEGER,

  -- Calculated
  mom_change INTEGER,               -- Month-over-month change
  mom_change_pct DECIMAL(6, 2),

```

```

yoy_change INTEGER,                -- Year-over-year change
yoy_change_pct DECIMAL(6, 2),

created_at TIMESTAMPTZ DEFAULT NOW(),

UNIQUE(business_id, snapshot_month)
);

CREATE INDEX idx_emp_trends_date ON employee_trends(snapshot_month DESC);

-- =====
-- EXECUTIVE DEPARTURES (PDL Recent Exec Departure)
-- =====
CREATE TABLE executive_departures (
  id UUID PRIMARY KEY DEFAULT gen_random_uuid(),
  business_id UUID REFERENCES businesses(id) ON DELETE CASCADE,

  executive_name VARCHAR(255),
  title VARCHAR(100),
  seniority_level VARCHAR(50),      -- 'c_suite', 'vp', 'director'

  departure_date DATE,
  departure_type VARCHAR(50),      -- 'resigned', 'terminated', 'retired', 'unknown'

  -- New Position (if known)
  new_company VARCHAR(255),
  new_title VARCHAR(100),

  data_source VARCHAR(50) DEFAULT 'pdl',
  raw_record JSONB,

  created_at TIMESTAMPTZ DEFAULT NOW(),

  UNIQUE(business_id, executive_name, departure_date)
);

CREATE INDEX idx_exec_departures_date ON executive_departures(departure_date DESC);

-- =====
-- PROPERTY DATA (ATTOM + Verisk)
-- =====
CREATE TABLE property_records (
  id UUID PRIMARY KEY DEFAULT gen_random_uuid(),
  business_id UUID REFERENCES businesses(id) ON DELETE CASCADE,

```

```

-- Property Identity
apn VARCHAR(50),
property_address VARCHAR(255),
city VARCHAR(100),
state VARCHAR(2),
zip VARCHAR(10),
county VARCHAR(100),

-- Assessor Parcel Number

-- Ownership
owner_type VARCHAR(50),
owner_name VARCHAR(255),

-- 'business_owned', 'owner_personal', 'leased'

-- Property Details
property_type VARCHAR(50),
year_built INTEGER,
building_sqft INTEGER,
lot_sqft INTEGER,

-- 'commercial', 'industrial', 'retail', 'office'

-- Valuation
assessed_value DECIMAL(15, 2),
market_value DECIMAL(15, 2),
avm_value DECIMAL(15, 2),
avm_confidence DECIMAL(3, 2),

-- Automated Valuation Model

-- Tax Status
tax_delinquent BOOLEAN DEFAULT FALSE,
tax_amount_due DECIMAL(12, 2),
last_tax_payment_date DATE,

-- Mortgage
mortgage_amount DECIMAL(15, 2),
mortgage_date DATE,
mortgage_lender VARCHAR(255),
loan_to_value DECIMAL(5, 2),

-- Lease Info (if applicable)
lease_expiration_date DATE,
monthly_rent DECIMAL(12, 2),

data_source VARCHAR(50),
raw_record JSONB,

-- 'attom', 'verisk'

created_at TIMESTAMPTZ DEFAULT NOW(),

```

```

updated_at TIMESTAMPTZ DEFAULT NOW(),

UNIQUE(apn)
);

CREATE INDEX idx_property_business ON property_records(business_id);
CREATE INDEX idx_property_tax ON property_records(tax_delinquent);

-- =====
-- BUILDING PERMITS (Bulity - Maintenance Signal)
-- =====
CREATE TABLE building_permits (
  id UUID PRIMARY KEY DEFAULT gen_random_uuid(),
  business_id UUID REFERENCES businesses(id) ON DELETE CASCADE,
  property_id UUID REFERENCES property_records(id),

  permit_number VARCHAR(100),
  permit_type VARCHAR(100),          -- 'renovation', 'repair', 'addition', 'new_construction'

  issue_date DATE,
  completion_date DATE,
  expiration_date DATE,

  estimated_cost DECIMAL(12, 2),
  actual_cost DECIMAL(12, 2),

  description TEXT,
  contractor_name VARCHAR(255),

  status VARCHAR(50),               -- 'issued', 'in_progress', 'completed', 'expired'
  jurisdiction VARCHAR(100),

  data_source VARCHAR(50) DEFAULT 'bulity',
  raw_record JSONB,

  created_at TIMESTAMPTZ DEFAULT NOW(),

  UNIQUE(permit_number, jurisdiction)
);

CREATE INDEX idx_permits_date ON building_permits(issue_date DESC);
CREATE INDEX idx_permits_business ON building_permits(business_id);

-- =====

```

```

-- FOOT TRAFFIC (SafeGraph / Advan)
-- =====
CREATE TABLE foot_traffic (
  id UUID PRIMARY KEY DEFAULT gen_random_uuid(),
  business_id UUID REFERENCES businesses(id) ON DELETE CASCADE,

  placekey VARCHAR(100),          -- SafeGraph PlaceKey

  -- Time Period
  period_start DATE NOT NULL,
  period_end DATE NOT NULL,
  period_type VARCHAR(20),        -- 'weekly', 'monthly'

  -- Traffic Metrics
  raw_visit_count INTEGER,
  normalized_visits INTEGER,
  unique_visitors INTEGER,

  -- Dwell Time
  median_dwell_minutes INTEGER,

  -- Demographics (if available)
  visitor_home_cbgs JSONB,        -- Census block groups of visitors

  -- Calculated Trends
  wow_change_pct DECIMAL(6, 2),   -- Week-over-week
  mom_change_pct DECIMAL(6, 2),   -- Month-over-month
  yoy_change_pct DECIMAL(6, 2),   -- Year-over-year

  data_source VARCHAR(50),        -- 'safegraph', 'advan'
  raw_record JSONB,

  created_at TIMESTAMPTZ DEFAULT NOW(),

  UNIQUE(business_id, period_start, period_type)
);

CREATE INDEX idx_traffic_date ON foot_traffic(period_start DESC);
CREATE INDEX idx_traffic_business ON foot_traffic(business_id);

-- =====
-- ONLINE REVIEWS & SENTIMENT (Yelp, Google)
-- =====
CREATE TABLE online_reviews (

```

```

id UUID PRIMARY KEY DEFAULT gen_random_uuid(),
business_id UUID REFERENCES businesses(id) ON DELETE CASCADE,

platform VARCHAR(20) NOT NULL,           -- 'yelp', 'google', 'facebook'
platform_business_id VARCHAR(255),       -- Yelp/Google business ID

-- Current Snapshot
snapshot_date DATE NOT NULL,
current_rating DECIMAL(2, 1),            -- 1.0-5.0
total_review_count INTEGER,

-- Review Velocity
reviews_last_30_days INTEGER,
reviews_last_90_days INTEGER,

-- Rating Trend
rating_30_days_ago DECIMAL(2, 1),
rating_90_days_ago DECIMAL(2, 1),
rating_trend VARCHAR(20),                -- 'improving', 'stable', 'declining'

-- Sentiment Analysis
positive_review_pct DECIMAL(5, 2),
negative_review_pct DECIMAL(5, 2),
recent_negative_spike BOOLEAN DEFAULT FALSE,

-- Price Level
price_level INTEGER,                     -- 1-4 ($-$$$$)

data_source VARCHAR(50),
raw_record JSONB,

created_at TIMESTAMPTZ DEFAULT NOW(),

UNIQUE(business_id, platform, snapshot_date)
);

CREATE INDEX idx_reviews_platform ON online_reviews(platform);
CREATE INDEX idx_reviews_rating ON online_reviews(current_rating);
CREATE INDEX idx_reviews_date ON online_reviews(snapshot_date DESC);

-- =====
-- SOCIAL MEDIA PRESENCE (Custom Scrapers)
-- =====
CREATE TABLE social_media_presence (

```

```

id UUID PRIMARY KEY DEFAULT gen_random_uuid(),
business_id UUID REFERENCES businesses(id) ON DELETE CASCADE,

platform VARCHAR(20) NOT NULL,           -- 'facebook', 'instagram', 'linkedin'
platform_url VARCHAR(500),
platform_id VARCHAR(255),

-- Snapshot
snapshot_date DATE NOT NULL,

-- Metrics
follower_count INTEGER,
following_count INTEGER,
total_posts INTEGER,

-- Activity
posts_last_30_days INTEGER,
posts_last_90_days INTEGER,
last_post_date DATE,
days_since_last_post INTEGER GENERATED ALWAYS AS (
    CURRENT_DATE - last_post_date
) STORED,

-- Engagement
avg_likes_per_post DECIMAL(10, 2),
avg_comments_per_post DECIMAL(10, 2),
engagement_rate DECIMAL(6, 4),

-- Trends
follower_trend VARCHAR(20),              -- 'growing', 'stable', 'declining'
activity_trend VARCHAR(20),              -- 'active', 'slowing', 'dormant', 'abandoned'

raw_record JSONB,

created_at TIMESTAMPTZ DEFAULT NOW(),

UNIQUE(business_id, platform, snapshot_date)
);

CREATE INDEX idx_social_platform ON social_media_presence(platform);
CREATE INDEX idx_social_activity ON social_media_presence(days_since_last_post);

-- =====
-- SEC FILINGS (For businesses with public parent/related entities)

```

```

-- =====
CREATE TABLE sec_filings (
  id UUID PRIMARY KEY DEFAULT gen_random_uuid(),
  business_id UUID REFERENCES businesses(id) ON DELETE CASCADE,

  cik VARCHAR(20),                -- SEC Central Index Key
  company_name VARCHAR(255),

  filing_type VARCHAR(20) NOT NULL,    -- '10-K', '10-Q', '8-K', 'Form4'
  filing_date DATE NOT NULL,
  accession_number VARCHAR(30),

  -- Extracted Signals
  going_concern_warning BOOLEAN DEFAULT FALSE,
  auditor_change BOOLEAN DEFAULT FALSE,
  ceo_departure BOOLEAN DEFAULT FALSE,
  cfo_departure BOOLEAN DEFAULT FALSE,
  material_weakness BOOLEAN DEFAULT FALSE,

  -- Form 4 Specifics
  insider_name VARCHAR(255),
  transaction_type VARCHAR(20),        -- 'buy', 'sell'
  shares_transacted INTEGER,
  transaction_value DECIMAL(15, 2),

  -- 8-K Event Types
  event_types JSONB,                  -- Array of 8-K item numbers

  filing_url VARCHAR(500),
  raw_text TEXT,

  created_at TIMESTAMPTZ DEFAULT NOW(),

  UNIQUE(accession_number)
);

CREATE INDEX idx_sec_filings_type ON sec_filings(filing_type);
CREATE INDEX idx_sec_filings_date ON sec_filings(filing_date DESC);

-- =====
-- JOB POSTINGS (WageScape - Hiring/Contraction Signal)
-- =====
CREATE TABLE job_postings (
  id UUID PRIMARY KEY DEFAULT gen_random_uuid(),

```



```

business_id UUID REFERENCES businesses(id) ON DELETE CASCADE,

-- Aggregated Monthly View
snapshot_month DATE NOT NULL,

active_postings_count INTEGER,
new_postings_count INTEGER,
filled_postings_count INTEGER,
expired_postings_count INTEGER,

-- Salary Data
avg_salary_offered DECIMAL(12, 2),
min_salary_offered DECIMAL(12, 2),
max_salary_offered DECIMAL(12, 2),

-- Trends
mom_posting_change INTEGER,
mom_posting_change_pct DECIMAL(6, 2),
yoy_posting_change INTEGER,
yoy_posting_change_pct DECIMAL(6, 2),

posting_trend VARCHAR(20),          -- 'hiring', 'stable', 'contracting'

data_source VARCHAR(50) DEFAULT 'wagescape',
raw_record JSONB,

created_at TIMESTAMPTZ DEFAULT NOW(),

UNIQUE(business_id, snapshot_month)
);

CREATE INDEX idx_job_postings_date ON job_postings(snapshot_month DESC);

-- =====
-- COMPETITIVE LANDSCAPE (PDL + Custom)
-- =====
CREATE TABLE local_competition (
  id UUID PRIMARY KEY DEFAULT gen_random_uuid(),
  business_id UUID REFERENCES businesses(id) ON DELETE CASCADE,

  snapshot_date DATE NOT NULL,

  -- Competitors within radius
  competitors_1_mile INTEGER,

```

```

competitors_3_mile INTEGER,
competitors_5_mile INTEGER,

-- New Competition
new_competitors_12mo INTEGER,          -- Opened in last 12 months within 3 miles
closed_competitors_12mo INTEGER,

-- Market Position
market_saturation_score DECIMAL(3, 2),  -- 0-1 scale

created_at TIMESTAMPTZ DEFAULT NOW(),

UNIQUE(business_id, snapshot_date)
);

-- =====
-- OWNER CONTACT INFO (Apollo + Whitepages)
-- =====
CREATE TABLE owner_contact_enrichment (
  id UUID PRIMARY KEY DEFAULT gen_random_uuid(),
  owner_id UUID REFERENCES business_owners(id) ON DELETE CASCADE,

  -- Apollo Data
  apollo_id VARCHAR(100),
  work_email VARCHAR(255),
  personal_email VARCHAR(255),
  work_phone VARCHAR(20),
  mobile_phone VARCHAR(20),
  linkedin_url VARCHAR(500),

  -- Email Verification
  email_verified BOOLEAN,
  email_verification_date DATE,
  email_deliverability VARCHAR(20),      -- 'deliverable', 'risky', 'undeliverable'

  -- Phone Verification
  phone_verified BOOLEAN,
  phone_type VARCHAR(20),                -- 'mobile', 'landline', 'voip'

  -- Whitepages Enrichment
  whitepages_verified_age INTEGER,
  associated_addresses JSONB,
  associated_relatives JSONB,

```

```
data_freshness_date DATE,  
  
created_at TIMESTAMPTZ DEFAULT NOW(),  
updated_at TIMESTAMPTZ DEFAULT NOW(),  
  
UNIQUE(owner_id)  
);
```

PART 3: THE 55 WEIGHTED FEATURES

Organized into **7 Signal Categories**. Each feature has:

- **Data Source:** Where it comes from
 - **Raw Signal:** What we're measuring
 - **Weight:** Contribution to final score (total = 100 points)
 - **Threshold Logic:** When it fires
-

CATEGORY 1: OWNER DEMOGRAPHICS & AGE (15 points total)

#	Feature	Source	Signal	Points	Logic
1	<code>owner_age</code>	Whitepages + PDL	Owner is nearing retirement	6	65+ = 6pts, 60-64 = 4pts, 55-59 = 2pts
2	<code>years_ownership</code>	PDL + SOS	Long ownership = burnout	3	25+ yrs = 3pts, 20-24 = 2pts, 15-19 = 1pt
3	<code>no_successor_identified</code>	PDL (no family in biz)	No clear successor	3	No family members in exec roles = 3pts
4	<code>owner_retirement_proximity</code>	Calculated	Years to 65	3	<2 yrs = 3pts, 2-5 yrs = 2pts, 5-10 yrs = 1pt

CATEGORY 2: OWNER LIFE EVENTS (18 points total)

#	Feature	Source	Signal	Points	Logic
5	recent_divorce	UniCourt	Divorce filed/finalized	6	Within 12mo = 6pts, 12-24mo = 4pts, 24-36mo = 2pts
6	recent_spouse_death	Obituaries + Public Records	Widowed	5	Within 12mo = 5pts, 12-24mo = 3pts
7	partnership_dissolution	UniCourt + SOS	Business partnership split	5	Within 12mo = 5pts, 12-24mo = 3pts
8	health_event_proxy	Court records + Social	Extended absence indicators	2	Court postponements, social silence combo = 2pts

CATEGORY 3: OWNER FINANCIAL DISTRESS (20 points total)

#	Feature	Source	Signal	Points	Logic
9	owner_personal_bankruptcy	UniCourt	Chapter 7 or 13 filed	8	Active/recent = 8pts, <3yrs discharged = 4pts
10	owner_tax_liens	County Recorder + UniCourt	IRS/State tax liens	5	Active = 5pts, satisfied <2yrs = 2pts
11	owner_civil_judgments	UniCourt	Judgments against owner	4	Active unsatisfied = 4pts, satisfied <2yrs = 2pts
12	owner_lien_count	UniCourt	Multiple liens = pattern	3	3+ liens = 3pts, 2 liens = 2pts, 1 lien = 1pt

CATEGORY 4: BUSINESS FINANCIAL DISTRESS (22 points total)

#	Feature	Source	Signal	Points	Logic
---	---------	--------	--------	--------	-------

13	pre_foreclosure_notice	ATTOM	Property in pre-foreclosure	7	Notice filed = 7pts
14	tax_delinquent_property	ATTOM	Property taxes overdue	5	>12mo delinquent = 5pts, 6-12mo = 3pts, <6mo = 1pt
15	business_bankruptcy_filed	UniCourt	Chapter 11 or 7 business BK	8	Filed = 8pts
16	ucc_filing_spike	ATTOM Recorder	Secured creditor filings	2	3+ new UCCs in 12mo = 2pts

CATEGORY 5: BUSINESS PERFORMANCE DECLINE (12 points total)

#	Feature	Source	Signal	Points	Logic
17	employee_count_decline	PDL Employee Trends	Workforce shrinking	4	>20% decline 12mo = 4pts, 10-20% = 2pts
18	executive_departure	PDL Exec Departures	Key exec left	3	C-suite <12mo = 3pts, VP = 2pts
19	paydex_score_low	D&B Scraper	Poor payment history	3	<40 = 3pts, 40-50 = 2pts, 50-60 = 1pt
20	failure_score_high	D&B Scraper	High failure probability	2	>1800 = 2pts, 1500-1800 = 1pt

CATEGORY 6: OPERATIONAL DECAY (8 points total)

#	Feature	Source	Signal	Points	Logic
21	no_permits_5_years	Builtly	Zero maintenance investment	3	No permits 5+ yrs = 3pts, 3-5 yrs = 2pts
22	foot_traffic_decline	SafeGraph/Avdon	Customers decreasing	3	>30% YoY decline = 3pts, 20-30% = 2pts, 10-20% = 1pt

2	job_postings_de	WageScape	Not hiring =	2	>50% decline YoY =
3	cline		stagnation		2pts, 25-50% = 1pt

CATEGORY 7: ONLINE REPUTATION & SENTIMENT (5 points total)

#	Feature	Source	Signal	Points	Logic
2	yelp_rating_decline	Yelp API	Rating dropping	2	Drop >0.5 in 6mo = 2pts, 0.3-0.5 = 1pt
4					
2	google_rating_decline	Google API	Rating dropping	2	Drop >0.5 in 6mo = 2pts, 0.3-0.5 = 1pt
5					
2	negative_review_spike	Yelp + Google	Sudden bad reviews	1	3+ negative reviews in 30 days = 1pt
6					

BONUS SIGNALS (Additional Features for Model Refinement)

These don't add to the 100-point base but provide additional discrimination:

#	Feature	Source	Signal	Use
27	social_media_abandoned	Custom Scraper	No posts 90+ days	Engagement decay
28	social_media_decline	Custom Scraper	Posting frequency dropped >50%	Activity slowdown
29	lease_expiration_near	Property Records	Lease ending <12mo	Timing trigger
30	loan_maturity_near	ATTOM Recorder	Balloon payment coming	Financial pressure
31	property_value_decline	ATTOM AVM	Property losing value	Asset erosion
32	new_competition_surge	PDL	3+ competitors opened nearby	Market pressure

33	industry_decline	NASDAQ benchmarks	Industry contracting	Macro factor
34	owner_age_times_tenure	Calculated	Age × Years owned	Compound burnout
35	multiple_distress_flags	Calculated	Count of active distress signals	Stacking indicator
36	revenue_decline_estimated	PDL + ConsumerEdge	Revenue proxy trending down	Financial health
37	entity_age_high	SOS	Business >25 years old	Succession risk
38	owner_multiple_businesses	PDL	Owner has other ventures	Potential consolidation
39	recent_zoning_change	County Records	Zoning changed nearby	Forced relocation risk
40	sec_going_concern	SEC Edgar	Going concern warning (for public subsidiaries)	Distress flag
41	insider_selling	SEC Edgar Form 4	Heavy insider sales	Bearish signal
42-55	Reserved for model feature expansion	Various	Additional signals identified through feedback loops	Continuous improvement

PART 4: ENSEMBLE MODEL ARCHITECTURE

Three models working together, weighted by performance:

MODEL 1: XGBoost (Gradient Boosting)

```
XGBOOST_CONFIG = {
  'objective': 'reg:squarederror', # Predicting score 0-100
  'max_depth': 7,
  'learning_rate': 0.03,
  'n_estimators': 300,
```

```
'subsample': 0.8,  
'colsample_bytree': 0.8,  
'reg_alpha': 0.1,  
'reg_lambda': 1.0,  
'min_child_weight': 3,  
'gamma': 0.1,  
'random_state': 42  
}  
# WHY: Best for capturing non-linear interactions (age × divorce × financial distress)  
# WEIGHT: 42%
```

MODEL 2: Random Forest

```
RANDOM_FOREST_CONFIG = {  
    'n_estimators': 250,  
    'max_depth': 12,  
    'min_samples_split': 8,  
    'min_samples_leaf': 4,  
    'max_features': 'sqrt',  
    'bootstrap': True,  
    'oob_score': True,  
    'random_state': 42,  
    'n_jobs': -1  
}  
# WHY: Robust to outliers, handles binary flags well, provides feature importance  
# WEIGHT: 33%
```

MODEL 3: Neural Network (MLP)

```
NEURAL_NET_CONFIG = {  
    'hidden_layer_sizes': (128, 64, 32),  
    'activation': 'relu',  
    'solver': 'adam',  
    'alpha': 0.001, # L2 regularization  
    'learning_rate': 'adaptive',  
    'learning_rate_init': 0.001,  
    'max_iter': 500,  
    'early_stopping': True,  
    'validation_fraction': 0.15,  
    'random_state': 42  
}  
# WHY: Captures temporal patterns (review velocity decay, social media abandonment)
```


WEIGHT: 25%

ENSEMBLE SCORING LOGIC

```
def calculate_ensemble_score(business_id: str) -> dict:
    """
    Calculate final score using weighted ensemble.
    Returns score 0-100 and tier assignment.
    """

    # Extract all features for this business
    features = extract_features(business_id) # Returns dict of 55 features
    feature_vector = np.array([features[key] for key in FEATURE_KEYS])

    # Scale for neural network
    feature_vector_scaled = scaler.transform(feature_vector.reshape(1, -1))

    # Get predictions from each model
    xgb_pred = xgboost_model.predict(feature_vector.reshape(1, -1))[0]
    rf_pred = random_forest_model.predict(feature_vector.reshape(1, -1))[0]
    nn_pred = neural_net_model.predict(feature_vector_scaled)[0]

    # Weighted ensemble
    ensemble_weights = {
        'xgboost': 0.42,
        'random_forest': 0.33,
        'neural_net': 0.25
    }

    final_score = (
        xgb_pred * ensemble_weights['xgboost'] +
        rf_pred * ensemble_weights['random_forest'] +
        nn_pred * ensemble_weights['neural_net']
    )

    # Clip to 0-100 range
    final_score = np.clip(final_score, 0, 100)

    # Assign tier based on score
    if final_score >= 75:
        tier = 'platinum' # 0-3 months - IMMEDIATE
        urgency = 'immediate'
    elif final_score >= 55:
```

```

    tier = 'gold'          # 3-6 months - HIGH PRIORITY
    urgency = 'high'
elif final_score >= 35:
    tier = 'silver'        # 6-12 months - NURTURE
    urgency = 'medium'
else:
    tier = 'unqualified'   # Not worth pursuing now
    urgency = 'low'

return {
    'business_id': business_id,
    'final_score': round(final_score, 2),
    'tier': tier,
    'urgency': urgency,
    'model_predictions': {
        'xgboost': round(xgb_pred, 2),
        'random_forest': round(rf_pred, 2),
        'neural_net': round(nn_pred, 2)
    },
    'top_contributing_signals': get_top_signals(features), # SHAP explainability
    'scored_at': datetime.utcnow().isoformat()
}

```

PART 5: TIER CLASSIFICATION

Tier	Score Range	Time to Sale	Action	Signal Profile
PLATINUM	75-100	0-3 months	Immediate outreach, high touch	Multiple critical signals stacking: Owner 65+ with divorce/health + financial distress + business decline. Or: Active bankruptcy + pre-foreclosure + partnership dissolution
GOLD	55-74	3-6 months	Priority outreach	2-3 strong signals: Owner 60+ with business decline. Or: Recent divorce + declining revenue. Or: Tax delinquency + employee shrinkage

SILVER	35-54	6-12 months	Nurture campaign	Early warning signals: Owner approaching retirement OR single financial distress flag OR operational decay starting
BLACK	0-34	12+ months or not selling	Monitor only	Few/no signals. Healthy business. Young owner. No distress.

PART 6: LLM EXPLANATION LAYER

After the ensemble scores, an LLM generates human-readable explanations:

```
def generate_lead_explanation(scored_lead: dict, features: dict) -> str:
    """
    Use LLM to generate natural language explanation of why this lead scored this way.
    """

    prompt = f"""
    You are an M&A analyst explaining why a business owner might be ready to sell.

    BUSINESS: {scored_lead['business_name']}
    SCORE: {scored_lead['final_score']}/100 ({scored_lead['tier'].upper()} tier)
    URGENCY: {scored_lead['urgency']} ({get_urgency_timeframe(scored_lead['urgency'])})

    TOP CONTRIBUTING FACTORS:
    {format_top_signals(scored_lead['top_contributing_signals'])}

    FULL SIGNAL PROFILE:
    - Owner Age: {features['owner_age']} years old
    - Years Ownership: {features['years_ownership']}
    - Recent Divorce: {features['recent_divorce']}
    - Personal Bankruptcy: {features['owner_personal_bankruptcy']}
    - Business Pre-Foreclosure: {features['pre_foreclosure_notice']}
    - Employee Trend: {features['employee_count_decline']}% decline
    - PAYDEX Score: {features['paydex_score_low']}
    - Online Rating Trend: {features['yelp_rating_decline']}

    Write a 2-3 sentence explanation for a business broker about why this owner
    is likely to sell within the indicated timeframe. Be specific about which
    signals are most compelling. Do NOT mention scores or algorithms - speak
    naturally about the business situation.
```

"""

```
explanation = llm.generate(prompt)
return explanation
```

Example Output:

```
# "This 67-year-old manufacturing company owner recently finalized a divorce
# and has two active tax liens totaling $180,000. Combined with a 25% workforce
# reduction over the past year and no building permits filed since 2019, this
# business shows strong indicators of an owner preparing to exit within 90 days."
```

PART 7: 3-STAGE FEEDBACK LOOP EXPLANATION

The 3-Stage Feedback Loop captures detailed outcomes at three critical stages and teaches the model: outreach attempts, calendar booking, deal closing. All data flows into the training system to continuously improve model accuracy.

Stage 1: Lead Outreach → Booked or Not Booked

Outcome Options:

Booked - Appointment successfully scheduled

Optional notes field

Proceeds to Stage 2

Training data: outcome_quality_score = 0.8

Not Booked - No Answer - Three call attempts completed with no contact

Detailed tracking: Attempt 1 (Tuesday 10 AM), Attempt 2 (Friday 10 AM), Attempt 3 (Monday 2 PM)

Lead marked inactive

Automatic retry scheduled in ninety days

Stays with assigned broker

Training data: outcome_quality_score = 0.1

Not Booked - Not Interested - Explicit rejection from owner

Required open text field for specific reason (e.g., "Already working with another broker", "Business doing too well to sell", "Not planning to sell within five years")

Lead marked inactive

Training data: outcome_quality_score = 0.2

Not Booked - DNC (Do Not Contact) - Owner requests no further contact

Lead permanently removed from outreach

Retained in database for compliance records

Training data: outcome_quality_score = 0.0

Not Booked - Future Interest - Owner provides callback timeframe

AI agent or broker captures specific callback date from owner

Lead remains active/pending on broker dashboard with callback date flagged

NOT marked inactive

System automatically schedules follow-up at specified time

Broker or AI reaches out again at callback date

Training data: outcome_quality_score = 0.6

Stage 2: Booked → Closed or Not Closed

After broker meets with prospect:

Closed - Won - Contract successfully signed

Signature obtained on Broker Contract

Commission amount entered (only after actual closing, not at meeting)

Full details logged for reporting

Training data: outcome_quality_score = 1.0

Closed - Lost - Deal did not proceed

Required open text field for reason (e.g., "Valuation expectations misaligned", "Decided not to sell after all", "Chose different broker", "Timing not right for sale")

Training data: outcome_quality_score = 0.3

Future - Pending - Prospect needs more time to decide

Broker enters note with expected timeline (e.g., "Will be ready in thirty days")

Lead stays active on broker dashboard

Broker continues receiving new weekly lead batches

Broker responsible for following up at appropriate time

No training data added until final outcome

Stage 3: Contract Signed → Deal Closed

On/After Closing Day

Complete/Deal Closed/Funded - Archive Lead

Transaction completed, closing docs signed and deal has been funded

Commission Amount Entered

Any additional details Broker wants to log for reporting or historical purposes
After close out “Lead” tag is changed to “Previous” and dropped into the Client database in the CRM/Dashboard under respective tab for later follow up or use

3-STAGE FEEDBACK LOOP ARCHITECTURE:

STAGE 1: OUTREACH OUTCOME

```
CREATE TABLE outreach_feedback (  
  id UUID PRIMARY KEY DEFAULT gen_random_uuid(),  
  lead_id UUID REFERENCES scored_leads(id),  
  
  -- Outcome  
  outcome VARCHAR(50) NOT NULL,          -- 'booked', 'callback_requested',  
  'not_interested', 'wrong_contact', 'no_response'  
  
  -- Reason (REQUIRED - from AI agent or broker)  
  outcome_reason TEXT NOT NULL,          -- "Owner said not ready yet, wants to wait until  
  lease ends"  
  
  -- Details  
  contact_method VARCHAR(20),            -- 'email', 'phone', 'ai_call'  
  contact_attempts INTEGER,  
  contact_date TIMESTAMPTZ,  
  
  -- Who provided feedback  
  feedback_source VARCHAR(20),           -- 'ai_agent', 'broker'  
  broker_id UUID,  
  
  created_at TIMESTAMPTZ DEFAULT NOW()  
);
```

STAGE 2: MEETING/BOOKING OUTCOME

```
CREATE TABLE meeting_feedback (  
  id UUID PRIMARY KEY DEFAULT gen_random_uuid(),  
  lead_id UUID REFERENCES scored_leads(id),  
  outreach_id UUID REFERENCES outreach_feedback(id),  
  
  -- Meeting Details  
  meeting_date TIMESTAMPTZ,  
  meeting_type VARCHAR(30),              -- 'discovery_call', 'valuation_presentation',  
  'listing_agreement'
```

```

-- Outcome
outcome VARCHAR(50) NOT NULL,          -- 'listing_signed', 'needs_follow_up',
'not_ready_now', 'declined'

-- Reason (REQUIRED)
outcome_reason TEXT NOT NULL,          -- "Owner wants to wait 6 months to get books
in order"

-- Seller Readiness Assessment
seller_motivation_level INTEGER,        -- 1-10 scale
estimated_timeline VARCHAR(30),        -- '0-3mo', '3-6mo', '6-12mo', '12mo+'

-- What changed from our prediction?
prediction_accuracy_notes TEXT,         -- "We predicted 0-3mo but owner says 6-12mo
because..."

broker_id UUID,
created_at TIMESTAMPTZ DEFAULT NOW()
);

```

STAGE 3: DEAL CLOSURE OUTCOME

```

CREATE TABLE deal_feedback (
  id UUID PRIMARY KEY DEFAULT gen_random_uuid(),
  lead_id UUID REFERENCES scored_leads(id),
  meeting_id UUID REFERENCES meeting_feedback(id),

  -- Deal Details
  listing_date DATE,
  sale_date DATE,
  days_to_close INTEGER,

  -- Final Outcome
  outcome VARCHAR(50) NOT NULL,        -- 'sold', 'listing_expired', 'withdrawn',
  'still_active'

  -- Sale Details
  asking_price DECIMAL(15, 2),
  sale_price DECIMAL(15, 2),
  sale_to_ask_ratio DECIMAL(5, 4),

  -- Post-Mortem

```

```

    why_they_sold TEXT,                -- Final confirmed reason for sale
    primary_motivation VARCHAR(50),    -- 'retirement', 'health', 'financial_distress',
    'burnout', 'opportunity', 'other'

    -- Model Accuracy
    original_score DECIMAL(5, 2),
    original_tier VARCHAR(20),
    actual_days_to_sale INTEGER,        -- Compare to predicted timeframe

    broker_id UUID,
    created_at TIMESTAMPTZ DEFAULT NOW()
);

```

FEEDBACK → MODEL RETRAINING

```

def prepare_training_data_from_feedback():
    """
    Convert closed feedback loops into training examples.
    """

    query = """
    SELECT
        sl.id,
        sl.feature_vector,
        sl.original_score,
        sl.tier,
        df.outcome,
        df.actual_days_to_sale,
        df.primary_motivation,
        mf.estimated_timeline,
        of.outcome as outreach_outcome
    FROM scored_leads sl
    JOIN deal_feedback df ON sl.id = df.lead_id
    LEFT JOIN meeting_feedback mf ON sl.id = mf.lead_id
    LEFT JOIN outreach_feedback of ON sl.id = of.lead_id
    WHERE df.outcome IN ('sold', 'listing_expired', 'withdrawn')
    AND df.created_at > NOW() - INTERVAL '12 months'
    """

    # Calculate actual score based on outcome
    for row in results:
        if row.outcome == 'sold':
            if row.actual_days_to_sale <= 90:

```



```

        actual_score = 85 + (90 - row.actual_days_to_sale) / 6 # 85-100
    elif row.actual_days_to_sale <= 180:
        actual_score = 60 + (180 - row.actual_days_to_sale) / 3.6 # 60-85
    elif row.actual_days_to_sale <= 365:
        actual_score = 35 + (365 - row.actual_days_to_sale) / 7.4 # 35-60
    else:
        actual_score = 35
    elif row.outcome == 'listing_expired':
        actual_score = 40 # They wanted to sell but couldn't
    else: # withdrawn
        actual_score = 20 # Changed mind

    training_examples.append({
        'features': row.feature_vector,
        'target': actual_score
    })

return training_examples

```

PART 8: DATABASE TABLES FOR ML PIPELINE

```

-- =====
-- SCORED LEADS (Output of ML Pipeline)
-- =====
CREATE TABLE scored_leads (
    id UUID PRIMARY KEY DEFAULT gen_random_uuid(),
    business_id UUID REFERENCES businesses(id) ON DELETE CASCADE,

    -- Scoring
    final_score DECIMAL(5, 2) NOT NULL,          -- 0.00-100.00
    tier VARCHAR(20) NOT NULL,                    -- 'platinum', 'gold', 'silver', 'unqualified'
    urgency VARCHAR(20) NOT NULL,                 -- 'immediate', 'high', 'medium', 'low'
    predicted_sale_window VARCHAR(20),            -- '0-3mo', '3-6mo', '6-12mo', '12mo+'

    -- Model Details
    model_predictions JSONB NOT NULL,              -- {"xgboost": 78.5, "random_forest": 74.2,
    "neural_net": 80.1}
    model_weights JSONB NOT NULL,                 -- {"xgboost": 0.42, "random_forest": 0.33,
    "neural_net": 0.25}
    feature_vector JSONB NOT NULL,                -- All 55 features with values

    -- Explainability
    top_signals JSONB NOT NULL,                   -- Top 5 contributing factors with SHAP values

```

```

llm_explanation TEXT,                -- Natural language explanation

-- Assignment
assigned_to_broker UUID,
assigned_at TIMESTAMPTZ,
territory_id UUID,

-- Status
status VARCHAR(30) DEFAULT 'new',    -- 'new', 'assigned', 'contacted',
'meeting_scheduled', 'closed', 'dead'

-- Timestamps
scored_at TIMESTAMPTZ DEFAULT NOW(),
last_rescored_at TIMESTAMPTZ,
score_version INTEGER DEFAULT 1,

created_at TIMESTAMPTZ DEFAULT NOW(),
updated_at TIMESTAMPTZ DEFAULT NOW()
);

CREATE INDEX idx_scored_tier ON scored_leads(tier);
CREATE INDEX idx_scored_score ON scored_leads(final_score DESC);
CREATE INDEX idx_scored_status ON scored_leads(status);
CREATE INDEX idx_scored_broker ON scored_leads(assigned_to_broker);

-- =====
-- MODEL VERSIONS (Track All Trained Models)
-- =====
CREATE TABLE model_versions (
    id UUID PRIMARY KEY DEFAULT gen_random_uuid(),

    model_name VARCHAR(50) NOT NULL,    -- 'xgboost_v1', 'random_forest_v2',
    'neural_net_v1'
    model_type VARCHAR(30) NOT NULL,    -- 'xgboost', 'random_forest', 'neural_net'
    version INTEGER NOT NULL,

-- Storage
model_path TEXT NOT NULL,             -- S3/storage path to serialized model
scaler_path TEXT,                    -- Path to fitted scaler (for NN)

-- Training Metrics
training_samples INTEGER,
training_date TIMESTAMPTZ NOT NULL,
training_rmse DECIMAL(6, 4),

```

```

validation_rmse DECIMAL(6, 4),
validation_r2 DECIMAL(5, 4),
validation_mae DECIMAL(6, 4),

-- Classification Metrics
precision_platinum DECIMAL(5, 4),
recall_platinum DECIMAL(5, 4),
precision_gold DECIMAL(5, 4),
recall_gold DECIMAL(5, 4),

-- Status
is_active BOOLEAN DEFAULT FALSE,
is_archived BOOLEAN DEFAULT FALSE,

-- Ensemble Weight (for active models)
ensemble_weight DECIMAL(4, 3),          -- 0.000-1.000

created_at TIMESTAMPTZ DEFAULT NOW(),

UNIQUE(model_type, version)
);

-- =====
-- ENSEMBLE CONFIGURATION (Current Active Ensemble)
-- =====
CREATE TABLE ensemble_config (
  id UUID PRIMARY KEY DEFAULT gen_random_uuid(),

  config_name VARCHAR(100) DEFAULT 'production',

  -- Active Model References
  xgboost_model_id UUID REFERENCES model_versions(id),
  random_forest_model_id UUID REFERENCES model_versions(id),
  neural_net_model_id UUID REFERENCES model_versions(id),

  -- Weights
  xgboost_weight DECIMAL(4, 3) DEFAULT 0.420,
  random_forest_weight DECIMAL(4, 3) DEFAULT 0.330,
  neural_net_weight DECIMAL(4, 3) DEFAULT 0.250,

  -- Tier Thresholds
  platinum_threshold INTEGER DEFAULT 75,
  gold_threshold INTEGER DEFAULT 55,
  silver_threshold INTEGER DEFAULT 35,

```

```

-- Status
is_active BOOLEAN DEFAULT TRUE,
activated_at TIMESTAMPTZ DEFAULT NOW(),

-- Performance Tracking
last_7d_accuracy DECIMAL(5, 4),
last_30d_accuracy DECIMAL(5, 4),

created_at TIMESTAMPTZ DEFAULT NOW(),
updated_at TIMESTAMPTZ DEFAULT NOW()
);

-- =====
-- ML TRAINING DATA (Historical outcomes for retraining)
-- =====
CREATE TABLE ml_training_data (
  id UUID PRIMARY KEY DEFAULT gen_random_uuid(),

  -- Source
  scored_lead_id UUID REFERENCES scored_leads(id),
  business_id UUID REFERENCES businesses(id),

  -- Features at time of scoring
  feature_vector JSONB NOT NULL,
  feature_version INTEGER NOT NULL,

  -- Original Prediction
  predicted_score DECIMAL(5, 2),
  predicted_tier VARCHAR(20),

  -- Actual Outcome (ground truth)
  actual_outcome VARCHAR(50),          -- 'sold_0_3mo', 'sold_3_6mo', 'sold_6_12mo',
  'not_sold', 'unknown'
  actual_score DECIMAL(5, 2),          -- Calculated from outcome timing

  -- Outcome Details
  days_to_outcome INTEGER,             -- Days from scoring to sale/decision
  outcome_reason TEXT,

  -- Quality
  confidence_level VARCHAR(20),        -- 'high', 'medium', 'low' - how certain are we
  about this training example

```

```

    created_at TIMESTAMPTZ DEFAULT NOW()
);

CREATE INDEX idx_training_outcome ON ml_training_data(actual_outcome);
CREATE INDEX idx_training_date ON ml_training_data(created_at DESC);

-- =====
-- FEATURE IMPORTANCE TRACKING (From model training)
-- =====
CREATE TABLE feature_importance (
    id UUID PRIMARY KEY DEFAULT gen_random_uuid(),
    model_version_id UUID REFERENCES model_versions(id),

    feature_name VARCHAR(100) NOT NULL,
    importance_score DECIMAL(8, 6),          -- Raw importance
    importance_rank INTEGER,                 -- 1 = most important
    shap_mean DECIMAL(8, 6),                 -- Mean absolute SHAP value

    created_at TIMESTAMPTZ DEFAULT NOW(),

    UNIQUE(model_version_id, feature_name)
);

-- =====
-- SCORING EVENTS (Audit trail of all scores)
-- =====
CREATE TABLE scoring_events (
    id UUID PRIMARY KEY DEFAULT gen_random_uuid(),
    business_id UUID REFERENCES businesses(id),
    scored_lead_id UUID REFERENCES scored_leads(id),

    event_type VARCHAR(30) NOT NULL,         -- 'initial_score', 'rescore_scheduled',
    'rescore_event_triggered'
    trigger_reason VARCHAR(100),             -- 'new_data', 'bankruptcy_filed',
    'scheduled_rescore'

    -- Score Change
    previous_score DECIMAL(5, 2),
    new_score DECIMAL(5, 2),
    score_change DECIMAL(5, 2),

    previous_tier VARCHAR(20),
    new_tier VARCHAR(20),

```

```

-- Model Used
ensemble_config_id UUID REFERENCES ensemble_config(id),

created_at TIMESTAMPTZ DEFAULT NOW()
);

CREATE INDEX idx_scoring_events_business ON scoring_events(business_id);
CREATE INDEX idx_scoring_events_date ON scoring_events(created_at DESC);

```

PART 9: EVENT-DRIVEN RESCORING

The platform continuously monitors external data sources for new events that could change a lead's score. When significant events are detected (court filings, new liens, executive departures, negative review spikes), the system automatically triggers re-scoring. When critical signals fire, rescore immediately.

Example Re-Scoring Triggers (Just a sample list, not all inclusive and others exist):

- New personal bankruptcy filing for business owner
- New tax lien or judgment filed
- Major life event detected (divorce, death in family)
- Sudden negative review spike on Yelp or Google
- New SEC 8-K filing (material event)
- Going concern warning in SEC filings
- Insider selling activity detected
- New pre-foreclosure notice
- Significant rating drop (>1.0 stars)
- Obituary Death Notice
- Personal or Business credit issues
- Releasing of several staff members including upper level management
- Decreased sales or activity i.e. foot traffic or website visits
- County recorder filings transferring ownership (or parts thereof)

Inngest Event Triggers

```

RESCORE_TRIGGERS = {
  'bankruptcy_filed': {
    'tables': ['owner_financial_distress', 'business_financial_distress'],
    'condition': "distress_type LIKE '%bankruptcy%' AND filing_date > NOW() - INTERVAL '7
days'",
    'priority': 'critical'
  },
}

```

```

'pre_foreclosure_notice': {
  'tables': ['business_financial_distress'],
  'condition': "distress_type = 'pre_foreclosure' AND notice_date > NOW() - INTERVAL '7
days'",
  'priority': 'critical'
},
'divorce_filed': {
  'tables': ['owner_life_events'],
  'condition': "event_type = 'divorce' AND event_date > NOW() - INTERVAL '7 days'",
  'priority': 'high'
},
'executive_departure': {
  'tables': ['executive_departures'],
  'condition': "seniority_level = 'c_suite' AND departure_date > NOW() - INTERVAL '14
days'",
  'priority': 'high'
},
'tax_delinquency': {
  'tables': ['property_records'],
  'condition': "tax_delinquent = TRUE AND updated_at > NOW() - INTERVAL '7 days'",
  'priority': 'medium'
},
'rating_crash': {
  'tables': ['online_reviews'],
  'condition': "rating_trend = 'declining' AND current_rating < 3.0 AND snapshot_date >
NOW() - INTERVAL '7 days'",
  'priority': 'medium'
}
}

```

Broker Onboarding Flow

The onboarding process ensures brokers configure their lead parameters and make an informed decision about AI calling concierge activation before receiving their first set of leads.

Onboarding Steps

Step 1: Account Creation

- User selects Tier 1 or Tier 2 Subscription and signs up via Clerk or Supabase authentication
- Email verification required
- Tenant record created in database

Step 2: Payment & Subscription

- Stripe checkout (unless you can recommend something better)- user pays setup fee and monthly subscription is initiated via autopay unless user chooses to be invoiced monthly (if invoiced this payment reminder is sent out 5 days prior to Payment Date (cycle begins the day they sign up and runs for 30 days)
- Subscription status is set to active upon successful payment
- Setup fee: one-time charge for platform access
- Monthly recurring: subscription for chosen Tier

Step 3: Lead Parameter Configuration

- Broker defines targeting criteria via multi-step form:
 - **Geographic scope:** Target cities, states, or zip codes
 - **Business size filters:** Minimum and/or maximum annual revenue, employee count range
 - **Business maturity:** Minimum years in business
 - **Industry preferences:** Select industries to include or exclude from lead pool
- Parameters saved to tenant_parameters table
- Validation ensures at least one filter is configured

AI Calling Concierge/Lead Qualification Explanation (This information is given to the broker at onboarding and they are required to sign. The meat and potatoes of the info is also included on the website along with the rest of the content for the platform *site for the domain is www.dealflow.ai - this domain is my application home and will be tied to Vercel and SupaBase. Serves as front door for the application. There will be all the relevant app info as well as sign up and sign in buttons/links on the homepage)

Before choosing a tier, the broker is presented with a clear explanation for both the AI calling concierge feature that is included in Tier 2 Subscription so they're fully informed.

Explanation Presented to Broker:

"The AI Calling Concierge automates your lead follow-up process. Here is how it works:

Email First Outreach: When new leads hit your dashboard every Tuesday at six a.m., an automated warm-up email is sent immediately to each prospect. This email includes valuable documents from your document vault, a personalized video introduction, plus your personal booking link for self-scheduling.

Monitoring Period: The system monitors email engagement for the duration of the outreach cycle, tracking opens, clicks, and replies.

At the same time, the email correspondence is sent out a USPS postcard is also deployed to the business owner's home address and business address with a QR code for scheduling.

These two outreach forms are complementary, no matter which tier you decide to go with, however, the lead qualification outreach cycle that follows is only available on tier 2 subscriptions.

OUTREACH ATTEMPTS:

INSERT HERE THE OUTREACH SCHEDULE

- If no contact is made either by email or telephone after the the initial 30 day outreach cycle the system pauses outreach of 14 days and then picks up the next 30 day cycle again until we speak to the lead and either book them on your calendar or they tell us to they are not interested at which time we will deactivate the lead in the database, however, the lead is assigned to you for life, so if something happens and they change their mind in 6 months or a year from now and reach out the lead reactivates and is immediately placed on your dashboard and calendar.

Phone Calls:

- The first follow up phone call initiates 72 hours after the initial email has been sent which is the day that the lead hit your dashboard
- Follow up phone calls are made every 48 hours thereafter between the hours of 10 AM and 6 PM and intend to schedule that lead on your calendar. The outreach phone calls and emails are done simultaneously throughout the entire 30 day outreach.. At which time if there's no contact made the 14 day pause is made on outreach and then the 30 day cycle resumes.

Emails:

- First is immediate when it hits your dashboard
- Second occurs between the phone call follow ups
- If no engagement then emails continue to be sent once a week for the 30 day outreach period (seen above) until resolved

Direct Mail Outreach Postcard

- We make sure to hit all the communication channels so we increase the visibility and stay top of mind with our leads.
- Sent as soon as the lead hits your dashboard at the same time the warm email is dispatched
- Goes to both the Business address and the owners home address

- You can design yourself or choose from one of our custom designed templates

Automatic Outcome Tracking: The AI agent records detailed outcomes for every call (booked, no answer, not interested, do not contact, future interest) and attaches call recordings and transcripts to each prospect record. If an appointment is booked, the AI creates the booking in your calendar and sends confirmation emails to you and the prospect.

If You DO NOT sign up for Tier 2 this Feature is not available to you with the exception of the initial Warm Email Outreach and Direct Mail Postcards: You will be responsible for all outreach after the initial touchpoint, follow-up calls, booking appointments, and actioning leads with outcomes. This is crucial for our machine learning system to continue delivering high-quality leads tailored to your preferences.

You can upgrade/downgrade your Tier Subscription anytime:

- From your dashboard, navigate to the “Account Tab” and select “My Subscription”, then click “Change My Tier” - at that point you can use the drop down menu to select your desired Tier. Billing amounts and dates will auto adjust and will reflect immediately in your account.
- There are no refunds for partial month subscriptions if you cancel, your account will remain accessible until the last day of your billing cycle. If you up/downgrade mid cycle we will prorate your month and apply any changes (credits or amounts due) immediately.
- If you prefer to reach out via telephone/email/chat you are welcome to do so and one of our account managers will be happy to take care of your request - help is available 24 hours a day.
- If you have requests or suggestions we would love to hear from you. We will work diligently to assess the request and if it falls within our operational wheelhouse to provide/ bake into the system then we will do all we can to provide you with the functionality you require to make your job easier.

[☐] I have read the terms associated with my new subscription and understand that if choosing Tier 2, I am responsible for all Outcome Feedback Actions as these directly impact model training and therefore tracking is required for optimal lead quality. Any breach of these terms shall constitute misuse and the owner (Aligned Enterprises dba DealFlow Ai, their beneficiaries, successors and assigns) may at their discretion, not requiring prior notice to me (the user) may terminate said subscription with no obligation to renew. I/We _____ agree to abide by the aforementioned requirement for use of this product.

Signature _____

Role

Date _____

Database Updates:

- User selection saved to `tenants.ai_calling_concierge_enabled` field
- Onboarding progress updated: `ai_concierge_decision_made` set to `TRUE`
- If user selects a Tier 2 Subscription, all AI call queue records for this tenant are skipped in Inngest workflows

Step 5: Onboarding Complete

- Confirmation screen displayed: "Your dashboard is ready. First lead drop scheduled for Tuesday (provide date) at 6 a.m."
- Call-to-action button: "Explore Your Dashboard"
- `onboarding_complete` flag set to `TRUE` in both `tenants` and `onboarding_progress` tables

Step 6: Interactive Dashboard Tour

- Guided walkthrough triggered on first dashboard login
- Highlights key features: lead map, scoring details, document vault, calendar integration, AI settings
- Tour can be skipped or replayed from settings
- Completion tracked in `onboarding_progress.dashboard_tour_completed`

Technology Stack (Help Me Nail This Down)

Core Infrastructure

- **Frontend Framework:** Nextjs
- **Styling:**
- **Database:** Supabase or Neon
- **Authentication:** Supabase or Clerk
- **Payments:** Stripe (tentative)
- **Hosting:** Vercel for frontend and API routes with edge functions

Or Something Like:

- ATTOM API → Apache Airflow pipeline
- Enrichment via Clay + Clearbit + FullContact
- Voice agent on Bland.ai or Retell
- Orchestration through n8n or Make
- Custom dashboard in Retool or Bubble
- Data warehouse in Snowflake or BigQuery

Workflow Orchestration

- **Background Jobs:** Inngest for event-driven workflows with Next.js integration

- **Scheduling:** Inngest scheduled functions for cron jobs (lead drops, model retraining, data pulls)
- **Queue Management:** Built-in Inngest queuing with automatic retries and observability

AI & Automation

- **Voice Agent:** for intelligent phone outreach with real-time tool calling (Hume or Eleven Lab or similar along with Vapi, Aircall or the like)
- **Email Delivery:** for automated email sends and engagement tracking (perhaps Resend)
- **Calendar Booking:** Cal.com self-hosted instance on Vercel for appointment scheduling
- **Machine Learning:** Python ML service using XGBoost, Random Forest, and Neural Networks with scikit-learn
- **Meta Model:** Powered by Claude or OpenRouter
- **Ai Dashboard Broker Assistant:** Powered by Claude or OpenRouter again)

Data Sources

- SEE SECTION ON DATA SOURCES IN ABOVE SECTIONS

Monitoring & Logging

- **Error Tracking:** Sentry for application error monitoring
- **Analytics:** Vercel Analytics for performance metrics
- **Logging:** Axiom or LogDrain for centralized logging

PRICING AND TIER OFFERING OPTIONS

Tier 1: Data Feed Only- \$6,500 Set Up Fee & (\$2,400/month)

- 100 qualified leads monthly from ATTOM
- Our Dashboard with CRM integration

Tier 2: Full Intelligence Platform- \$8,500 & (\$5,500/month)

- Everything in Tier 1
- AI voice qualifier
- Multichannel orchestration
- Psychology profiling

BUILD TIERS ONE AND TWO TO LAUNCH FIRST THEN WE IMPLEMENT TIER THREE

Tier 3: White Glove Execution- \$12,500 Set Up Fee & (\$10,500/month + performance)

- Everything in Tier 2
- Your team manages campaigns
- Guaranteed appointments
- Revenue share on closed deals

Tier 1: Data Feed

- 60 Off Market pre-qualified leads monthly (Can request higher lead limits)
- Warm 1st Touch Outreach/Intro Email & Direct Mailer to Business Address and Owner Home Address
- CRM integration and Intelligent Dashboard
- Dealroom, Document Vault, Client Database, Digital Assistant (Ava- DealScout), Voice Notes, Calendar & Booking, Advanced Reporting w/ Close Rates/Commission/Value Tracker, Dotted (In-App Document Signing), Secure Doc Upload, Messaging and more.

Tier 2: Full Intelligence Platform

60 Off Market pre-qualified leads monthly (Can request higher lead limits)

- Warm 1st Touch Outreach/Intro Email & Direct Mailer to Business Address and Owner Home Address
- CRM integration and Intelligent Dashboard
- Dealroom, Document Vault, Client Database, Digital Assistant (Ava- DealScout), Voice Notes, Calendar & Booking, Advanced Reporting w/ Close Rates/Commission/Value Tracker, Dotted (In-App Document Signing), Secure Doc Upload, Messaging and more.
- AI Voice Agent Outreach - Lead Qualification and Appointment Booking System
- Enhanced Functionality via Psychological Profiling of All Leads using Ai to listen to and assess all recordings of phone calls/OCR of emails to obtain owner intent, voice inflections/behaviour cues/cadence ect to gauge not only the likely “sell factors” driving the business owner but pain points, what makes them tick, any personal hits they reveal in combination with other factors from when the lead was scored before assignment- give a whole new meaning to KYC!

- Optional payment reporting to business/personal credit reporting agencies (have your on time payments WORK FOR YOU, grow your business or personal credit while closing leads)

TIER 3- WHITE GLOVE EXECUTION

- Everything in Tier 2
 - Your team manages campaigns
 - Guaranteed appointments
 - Revenue share on closed deals
-

ADMIN DASHBOARD (Need your help fleshing this part out)

USER DASHBOARD

USER DASHBOARD SET UP

1. **HOME:** (SEE IN DEPTH EXPLANATION ABOVE AND IN DOCUMENT 3 CONJUNCTIVELY)
2. **PROSPECTS** (LEADS DROPPED TO DASHBOARD LAND HERE)
3. **MY CLIENTS** (ALL CONTACTS IMPORTED FROM OTHER CRM APPLICATIONS VIA API)
4. **ACTIVE DEALS** (KANBAN PIPELINE THAT MOVES THE NEW CLIENT THROUGH THE STAGES OF THE DEAL FROM SIGNING UNTIL CLOSING)

5. **CALENDAR** (ALL SCHEDULED APPTS WITH BUSINESS OWNERS)
6. **DEALROOM** (DIRECT MESSAGING BETWEEN BROKER AND OWNER(S), SECURE DOCUMENT SIGNING AND FORWARDING OR UPLOADING CLIENT SIDE OR BROKER SIDE WITH REAL TIME ALERTS FOR BROKER/USER SO OWNER GETS IMMEDIATE RESPONSES TO QUESTIONS OR ISSUES, ECT.)
7. **OUTREACH CONCIERGE** - THIS IS THE AUTOMATED OUTREACH AND LEAD QUALIFICATION SECTION WHERE THE USER CAN ADJUST THE SCRIPT THE AI AGENT USES, CAN SEE WHO WAS CONTACTED THAT DAY ALONG WITH LISTEN TO THE CALL RECORDING OF THE OUTREACH AND VIEW THE OUTCOME OF THE CALL OR EMAIL. ALONG WITH BEING INCLUDED ON THIS PORTION OF THE DASHBOARD FOR QUICK AND EASY VIEWING, ALL CALLS ARE RECORDED AND TRANSCRIBED AND ATTACHED AS AN ASSET TO THE LEAD IN THE SYSTEM SO THE BROKER/USER CAN NOT ONLY LISTEN TO CALLS BUT THEY CAN ALSO EXPORT THAT DATA ALONG WITH ALL OTHER MATERIALS SUCH AS APPOINTMENTS AND LEADS)
8. **DEAL ASSIST AI** - THIS IS A FLOATING WIDGET IN THE BOTTOM RIGHT CORNER OF THE SCREEN THAT IS VOICE AND CHAT ACTIVATED AND ACTS AS A PERSONAL ASSISTANT FOR THE USER/BROKER...THE AI HAS THE ABILITY TO ONLINE SEARCH, SEND AND READ OR RESPOND TO EMAILS, BREAK DOWN METRICS FOR THE USER, CALCULATE COMMISSIONS OR HELP THEM RESCHEDULE A MEETING OR EVEN ITERATE THEIR DAY FIRST THING IN THE MORNING AND EVEN WHIP UP A DOCUMENT AND SHOOT IT OVER TO AN OWNER FOR SIGNATURE.
9. **TASK MANAGER** - THE TASK MANAGER HOUSES ALL LEADS OR APPOINTMENTS THAT STILL REQUIRE THE USER TO "ACTION" THEM FOR THE FEEDBACK LOOP WITH A GENTLE REMINDER "ANY OUTSTANDING OUTREACHES WHICH NEED FOLLOW UP", IT ALSO ALLOWS THE USER TO SEE ANY DOCS NEEDING TO BE SENT OR REMINDERS FOR SITE VISITS AND THE LIKE.
10. **DOCUMENT VAULT** - THIS HOUSES ALL THE CONTRACTS, NDA'S, THE VALUE ADDING PDF DOCS THAT ARE ATTACHED TO THE INITIAL OUTREACH EMAIL SENT TO THE BUSINESS OWNER, SAMPLE LOI'S AND EVEN MARKETING COPY FOR PLACING ADS ONLINE. ALL DOCS ARE IMPORTABLE AND EXPORTABLE AS THE USER WISHES
11. **REPORTS AND METRICS** - THIS IS AN AREA WHICH THE USER CAN GET INCREDIBLY GRANULAR WITH THEIR STATS ON OUR PLATFORM BY TARGETING YTD COMMISSIONS, CLOSING RATIOS AS OPPOSED TO APPOINTMENTS MADE AND FROM WHICH COLOR GROUP OF THE LEAD

DISBURSEMENT EVERY WEEK AND THEY CAN CHOOSE TO VIEW THIS AS BAR, LINE OR PIE GRAPH OR STRAIGHT NUMBERS ONLY ON AN EXCEL SPREADSHEET.

12. **SETTINGS** - THIS IS WHERE THE USER CAN TOGGLE ON OR OFF THE AI OUTREACH, CHANGE LIGHT TO DARK MODE, PAUSE LEAD DISBURSEMENTS FOR THE NEXT 1-2 WEEKS, CHANGE THEIR ACCOUNT SETTINGS IN GENERAL.

FEATURE LIST EXPANDED:

1. AI CONCIERGE - AUTOMATED OUTREACH SYSTEM

- a. Feature can be “upgraded to or downgraded from” via the settings tab on the users dashboard
- b. For all other outreach details please see that portion of the document.
- c. All calls are recorded and transcribed then attached to the Lead in the User Dashboard as an asset that can be listened to, read or even exported at the Users leisure.
- d. There is a navigation link “Ai Concierge” that houses the ability to edit the body contents of the email being sent to the business owners, the calling script the Ai uses for outreach calls and the PDF attachments should the User wish to use their own docs they can upload any of their choosing. Again, the toggle for this feature is under the “Settings” page link in the navigation pane.

2. **The Leads will be housed under the navigation link/button “Prospects”**. The reason for this is because we also have a link to “Clients” should the User want to import any of their previous contacts or clients lists from google or excel as well as via API call if they are interested in utilizing our platform for their CRM needs.

- a. All leads now living under the “Prospects” tab will be in list form sectioned out by color (Platinum, Gold or Silver). Each of them will show the basic business info until and unless the User hovers over the business name at which point the contact info will show with last actions taken on the Prospect. Should they click on the business name it will expand the Prospect with a new card type window that pops up containing all the relative info for that Prospect including address, contact info, ownership info, last actions taken, the reasoning for the scoring and list placement ect.
- b. On the same Prospects page, there will be an instance of either Google or Apple Maps that has all the Prospects as corresponding color coded pins dropped on the map in the location of the business. Also expands on hover.

- c. Somewhere on the page will be a Google or Brave Browser Search Bar (embedded) where the User can do a quick search for any of the Prospects on their lists to get more info not included in the Prospect instance.
 - d. All Prospects are exportable
- 3. **On the Home Page of the User Dashboard-** the Navigation Pane will be vertical along the left hand side of the screen.
 - a. Home Page will include graphs, charts and breakdowns for a series of important metrics that way the User can see value as soon as they pop into the program.
 - b. The Deal Assist widget (Ava) will live in the bottom right hand corner of the screen floating that awakes on voice command or can be clicked into for the full chat input window. Powered by Anthropic Claude or OpenRouter.
 - c. There will be quick glance areas that show things like upcoming appointments, Prospect/Leads that need to be “Actioned”, a Task To Do List, and maybe a quick list view of the most recent lead disbursement to their dashboard with the explanations for the scoring so they can see what just came through and why.
- 4. **My Appointments**
 - a. This is the Calendar and appointment area that will show a forward looking 30 day calendar and then a list of the upcoming appointments and tasks for the week.
 - b. There will also be the option to connect their own google calendar instance rather than using the Cal.com instance we self host. This is up to the User. (Unless you feel there is a better way to or program to use to do this)
 - c. Appointments are all auto-confirmed at booking with an sms or email to all parties attending and then a secondary reminder email or sms the day before the appointment is to take place. All rescheduled appointments will show up as such alerting the broker of the change and once again sending another confirmation email with all the new appointment details.
 - d. All appointments can be exported.
- 5. **Pipeline**
 - a. This is a kanban style drag and drop Deal Pipeline where the user can move the Business Owner/Prospect through the 7 stages of the deal pipeline as they are making their way through the sale of their business AND/OR maybe we set something up on the lead/prospect with like a drop down of some kind where if the user or ai agent can change the deal stage from there and have the lead/prospect automatically flip into the chosen “stage”. Starting with

Contract Signing for the Broker to Represent them during the Sale and ending with Deal Funded/Transition Complete. Each of the Prospects will include a Quick Hit button that will take them from Prospect to a Client, Move To Deal Room, Enter Pipeline, Action This Outreach, Send Email, Send SMS.

6. Deal Room

- a. The Deal Room is a place for the Business Owner and the Broker to send secure messages back and forth, sign and notarize documents, client side or broker side can upload required docs, send the requested docs such as financials or NDA's, ect. The messaging is in real time and will ping the Broker as soon as a message comes through. Prospects enter the Deal Room once they become Clients and have entered the active Deal Pipeline. User can move Clients or Prospects around manually otherwise it will auto follow the flow listed above.

7. Deal Assist

- a. Deal Assist is an Ai Personal Assistant that not only has access to a customizable knowledge base (editable by the User but is initially set by the Admin and will be standard prompt across the board), the Ai will also have access to the internet for searches, can make, cancel, reschedule appointments, send emails to Clients or Prospects or even respond to ones received, it can hands free take notes on the fly or add tasks to the To Do List and even give them directions to a location, run reports or just give them a rundown of their day at the close or perhaps in the morning the User wants to know what is on the agenda, it will give specifics or just highlights depending on user preference. It can act as a Senior M&B Analyst and at the Users request provide in depth chart style reviews of the weeks new leads along with a deep dive report into the key metrics for the grouping to look for trends. And so much more!

8. Report Metrics

- a. Reports will be tracking all commissions made by the User/Broker, Conversion Rates for each of the Color Groups, Gets as granular as specific areas seeing the most action down to the zipcode, how they stack up to other users on the platform such as a leaderboard for appointments that turned into an Active Client from a Prospect and several others that apply. All reports are printable and savable as PDF.

9. Settings

- a. The Settings tab will house the toggle for Light Mode to Dark Mode, Export All Data, Terminate Subscription, Log Out, API info for the import or export of the Users Clients/Prospects ect., and a “How To Use The Platform” video guide along with a message admin button that goes right to the Admin Dashboard for response as well as a contact us button.

10. Document Vault

- a. The Document Vault houses our premade PDF EBook, the Checklist, Contract specific to the Broker/User, sample NDA's and LOL's. The User may upload any of their own docs to the Vault and has the ability to attach them to the Deal Room, or an Email message to a Client or Prospect at any time. All PDF's in the vault can be modified and edited by the User as well as Downloaded/Exported.

11. Task Manager

- a. The task manager has a Tick Tick style ToDo List and Reminder instance (please see the domain www.ticktick.com for specifics this tab should include).